

Top 10 Supercomputers and CPU Specifications

#1: **El Capitan** (United States)

The world's fastest supercomputer consists of 43,808 AMD fourth-generation EPYC "Genoa" 24-core 1.8 GHz CPUs (1,051,392 cores).

#2: **Frontier** (United States)

Uses 9,472 AMD EPYC 7713 64-core 2 GHz CPUs, which come to a total of 606,208 cores. Frontier reached a maximum of 1.35 exaflops using standard double-precision arithmetic.

#3: **Aurora** (United States)

Aurora uses 21,248 Intel Xeon CPU Max Series CPUs (each running at 2.4 GHz) which also use 64-bit architecture. It's famous for being the world's first exascale supercomputer, reaching a performance peak of 1.012 exaflops.

#4: **JUPITER Booster** (Germany)

The first and fastest non-American based supercomputer on the list, the JUPITER Booster uses Nvidia Grace ARM CPUs, with 72 Neoverse V2 cores/CPU. Each CPU has a clock rate of 3.3 GHz. The JUPITER is Europe's first exascale supercomputer, reaching a performance peak of 1 exaflop.

#5: **Eagle** (United States)

The Microsoft Azure Eagle uses Intel Xeon Platinum 8480C 48-core 2 GHz CPUs, with a total core count of 1,123,200. The Eagle reached a peak performance of 561 petaflops (HPL benchmark).

#6: **HPC6** (Italy)

The HPC6 uses AMD EPYC 3rd Gen (64-core, 2GHz) CPUs. The computer uses 3,472 CPUs, with 221,808 cores total. The HPC6 reached a peak performance of 606 petaflops.

#7: Supercomputer Fugaku (Japan)

Uses Fujitsu A64FX CPUs (ARM-based), normal mode clock rate of 2.0 GHz with 48 cores/cpu (7,630,848 total cores). Fugaku reached a max performance of 442 petaflops.

#8: Alps (Switzerland)

Uses Nvidia Grace ARM CPUs, with 72 cores per CPU. The CPUs get up to 3.1 GHz with normal performance. In the HPL benchmark, it reached 434.9 petaflops. This is the second supercomputer on this list that uses the Nvidia Grace series CPUs.

#9: LUMI (Finland)

LUMI uses the AMD EPYC 3rd Gen “Milan” CPUs (64-core, 2 GHz), with a total of 11,912 cores. This is the fourth supercomputer on this list to use an AMD CPU. The HPL benchmark was 370.7 petaflops sustained performance and a 531.51 petaflops theoretical peak.

#10: Leonardo (Italy)

The Leonardo uses Intel Xeon Platinum 8358 (Ice Lake) CPUs, with 32 cores per CPU, and a clock speed of 2.6 GHz. The HPL benchmark was 174.7 petaflops in November 2022, but the current metric is 241.2 petaflops.

CPU Representation

Total CPUs by manufacturer:

AMD	4
Intel	3
Nvidia	2
Fujitsu	1

Overall, Intel CPUs are not underrepresented in the top 500; the most popular CPU designer is Intel. After some research, I found that trends tend to favor AMD more in terms of more powerful and future supercomputer CPU usage. While Intel still holds the majority of the top 500 supercomputers, AMD is dominating by holding the few highest-performing computers. In other words, AMD is less popular but more powerful, Intel is more popular but less powerful (not generally, just in terms of supercomputer CPU implementation).

In researching for this assignment, I learned that Nvidia does not only design GPUs but CPUs as well. This is a recent change announced in 2021 at the GPU Technology Conference. Nvidia is a fabless company, meaning they only design the GPU and CPUs and outsource the manufacturing to foundries, mainly TSMC and sometimes Samsung.

The rivalry between Intel and AMD seems to be as old as time and stretches across many domains in the computing market. What stood out to me, however, is that the two companies' competition has been vital to creating CPU architectures advanced enough to support the world's most powerful supercomputers. With Nvidia being added into the mix recently, I hope to see CPU and general computing development advanced even further and faster.

Sources

<https://top500.org/lists/top500/2025/11/>

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